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1	June 30, 2010	Added clauses for SAS IEDs and done corrections in various tables such as regrouping of non-trip alarm signals,deleted duplicated alarm litra and corrected alarm indications
2		
3		

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SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA)

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

This Engineering Standard sets the minimum technical requirements for design, engineering and installation of the Supervisory Control and Data Acquisition (SCADA) system to remotely control and monitor Saudi Electricity Company (SEC) substations including grid stations and BSPs.

2.0 FUNCTION

2.1 The SCADA system shall be able to monitor and transmit the following information to the SEC Power Control Center (PCC) master station and/or any other designated control center :

2.1.1 Equipment status or indication such as circuit breaker status (open or closed) and status of various alarm points (normal/alarm).

2.1.2 Analog data such as line megawatts or megavars, bus kilovolts and transformer tap position.

2.1.3 Instantaneous Energy Meter values in pulse, digital or analog format. The Digital Counter Values shall be capable of operating in the following modes:

- Remotely Frozen With Reset
- Remotely Frozen Without Reset
- Locally Frozen With Reset
- Locally Frozen Without Reset

2.2 The SCADA system shall be able to respond to control commands from the Master Station and/or any other designated Control Center and relay these commands to the substation equipment such as :

2.2.1 Open or close of a circuit breaker.

2.2.2 Switch the control mode of substation equipment.

2.2.3 Select transformer tap changer control scheme and raise or lower the tap position of a transformer.

2.3 SCADA system shall be locally programmable to independently initiate, conduct, and terminate Closed Loop Control (CLC), trigger automatic control of substation equipment based on telemetered real time analog and/or digital values. The provision of Automatic Controlling shall be in accordance with IEC standard (Logic CAD; IEC 61131-3); Equivalent standard from ANSI can be accepted.

3.0 SCADA EQUIPMENT/MATERIALS

3.1 General

All items referred and including items not covered, in this standard shall be in accordance with the latest revision/amendments of applicable SEC Standards & Specifications and to the latest edition of applicable International Industry codes and standards.

3.2 Remote Terminal Unit

3.2.1 The Remote Terminal Unit (RTU) shall comply with SEC Transmision Material Standard Specification no. 38-TMSS-03.

3.2.2 The RTU Local Area Network (LAN) shall have adequate protection from noise that is inherent at the substations.

3.3 SAS Control and Protection IEDs

3.3.1 SAS IEDs shall be capable to initiate all alarm as per latest approved TMSS and shall be communicated to SEC Power Control Center (PCC).

3.4 Transducers

3.4.1 General Requirements

a. Transducers shall be provided to convert primary power system measured quantities into a form, which can be accepted by the RTU.

b. Transducers shall comply with IEC 60688. They shall be of the numeric or solid-state design with no moving parts.

c. Transducer outputs shall be dedicated to SCADA for all voltage levels above 34.5kV. For 34.5kV system and below same shall be provided if specified in Project Technical Specification (PTS).

d. Transducer inputs and outputs shall be galvanically isolated from each other and from the auxiliary power supply.

e. Transducers shall be capable of providing a load independent DC output current into a load of up to 1000 ohms. The maximum ripple in the DC output signal shall be 1 %. The transducer burden shall not exceed 2 VA.

3.4.2 Active / Reactive Power Transducers (MW / MVAR)

The input of all active/reactive power transducers and the corresponding PT/CT ratios shall be such that the full scale output represents a minimum of 120 % of the maximum line or transformer capacity. The transducer shall be able to accurately measure the MW and MVAR quantities of a 3-phase

unbalanced line into independent DC current varying in proportion to the measured value of the active or reactive power.

No. of Elements	:	3 (Minimum)
Nominal Potential Input	:	Per PT secondary requirement
Nominal Current Input	:	1 or 5 A
Rated Output Signal	:	± 10 Milliamps DC (bipolar)
Wave Form	:	Sinusoidal
Accuracy	:	0.25 % or better per IEC (60688)

3.4.3 Voltage Transducer

The voltage transducer is designed to convert an AC voltage into a load independent DC signal. The input range of all voltage transducers and the corresponding PT ratios shall be such that the full scale output of the transducer represents a line or bus voltage of at least 120 % of the nominal line or bus voltage.

Nominal Potential Input	:	Per PT secondary requirement
Rated Output	:	4 – 20 Milliamps DC (unipolar)
Accuracy	:	0.25 % or better per IEC (60688)

3.4.4 Current Transducer

The input range of all current transducers and the corresponding CT ratios shall be such that the full scale output of the transducer represents a line current of at least 110% of the nominal line current.

Nominal Current Input	:	1 or 5 Amperes, 60 Hz
Rated Output	:	± 10 Milliamps DC (unipolar)
Accuracy	:	0.25% or Better

3.4.5 Electronic Tap-Position Transducer

Tap Position	:	Lowest - Highest
Rated Output	:	4 – 20 Milliamps DC (unipolar)
Accuracy	:	0.25% or Better

3.4.6 Frequency Transducer

Input Range	:	55 - 65 Hz
Output Range	:	4 – 20 Milliamps DC (unipolar)
Accuracy	:	0.01% or Better in range of 58-62 Hz

3.5 Equipment Interface

3.5.1 The voltage and current ratings of the RTU I/O modules specified in 38-TMSS-03 shall never be exceeded in any substation equipment interface.

Status and alarm inputs shall be dry contacts (no external voltage) dedicated to SCADA.

3.5.2 Intelligent Electronic Device (IED) interface shall conform to ANSI/TIA-232-F "Interface Between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange".

3.5.3 Closed Loop Control (CLC) interface shall also conform with ANSI/TIA-232-F.

3.6 Control Interface

Remote control interface shall adhere to the following convention :

Table : 27-1

SCADA Control Output	Close	Open (Trip)
Field Control Option Equivalent	Close	Open
	On	Off
	Auto	Manual
	Start	Stop
	Raise	Lower
	In	Out

3.7 Status and Alarms Contacts

3.7.1 The condition that exists on the closure of a contact for status shall be noted in the Function Tabulation Sheets. The following convention shall be followed :

Table : 27-2

Input Type	State	Input Contact Status
Alarms	Normal Abnormal	Closed Open
Circuit Breaker Status	Closed Open	Closed Open
Breaker Control Mode	Local Remote	Open Closed
ABTS	In Out	Closed Open
Bus Differential Protection	In Out	Closed Open
Breaker Failure Protection	In Out	Closed Open

Table 27-2 (Continued)

Input Type	State	Input Contact Status
Transformer RTCC Status	Manual (AVR Out) Auto (AVR In)	Closed Open
	Supervisory Local	Closed Open
	Master	Closed
	Parallel Independent	Closed Open
	Tap Position	Analog Indication

3.7.2 Where alarms are combined into a single point, a serial combination of closed contacts (when in normal state) shall be provided for each alarm.

3.7.3 Cable runs between monitoring devices and the RTU shall be radially paired. Pairs used for runs to the RTU shall be in the same cable.

3.7.4 Alarm contacts to the RTU shall not be affected by the annunciation operation in any way. These shall not be latched by the annunciator and shall not change state when an annunciator test is performed.

3.8 SCADA Cables

SCADA analog input cables shall be shielded/sheathed, twisted pair cables. SCADA status and control cables shall be overall group shielded and comply with the requirements of SEC Material Standard Specification 11-TMSS-11

3.9 Conduits and Cable Trays

Conduits and cable trays shall be in accordance with the requirements of TES-P-119.20 and 24-TMSS-01.

4.0 INSTALLATION OF SCADA EQUIPMENT

4.1 Remote Terminal Unit

4.1.1 The design shall take into consideration the type of RTU enclosure specified in 38-TMSS-03 data schedule.

4.1.2 Type I enclosure (with Field Interface Cabinet) shall be in accordance with 38-TMSS-03.

4.1.3 For type II enclosures (without Field Interface Cabinet):

a. I/O controller enclosures adjacent to the relay/control panels are preferred where it is practically possible.

b. The design shall systematically group SCADA points originating from a single row of adjacent relay/control panels then wired to a common location to form the I/O controller enclosure.

c. The number of enclosures shall depend on the number of rows of adjacent relay/control panels or locations where SCADA point sources are concentrated.

d. Enclosures shall be able to accommodate all the I/O controllers required for a given row of adjacent relay/control panels.

e. Enclosures shall be properly labeled showing enclosure number, and the addresses of I/O controllers contained.

4.1.4 The RTU power supply requirements, which shall consist of primary DC from station battery and 127V, 60 Hz AC back up shall be adapted.

4.2 SAS IEDs

4.2.1 SAS IEDs shall be as per relevant standard or Project SOW/PTS**.**

4.3 Transducers

4.3.1 The control panel, metering panel or low voltage compartment of 34.5 kV / 33 kV / 13.8 kV / 11 kV switchgear/RTU panel are the preferred locations for transducers where practicable.

4.3.2 Transducers shall be labeled to correspond with SCADA functional diagrams.

4.3.3 Test switches (similar in design and function to Type FT1 test switches) shall be provided in metering circuits to allow transducer installation, testing and maintenance without interruption to other circuits.

4.4 Laying of SCADA Cables

4.4.1 SCADA Cables shall be kept away from high voltage cables as far as possible.

4.4.2 SCADA analog, status and control cables shall be grounded per TES-P-119.10.

4.4.3 SCADA wiring cable tails shall be bonded in such a way that each wire could be traced back to its associated cable.

4.4.4 Cable ties shall be used at all equipment to support the weight of the cables.

4.4.5 Each core of the control and instrumentation cables shall be identified with numbered ferrules. The ferrules shall be made of insulating materials. A wire list shall be prepared during the design stage. The list shall include the wire size, color, function, origin, destination at the terminal blocks and the ferrule labels.

4.4.6 Cable identification number shall consist of equipment identification numbers where the cable terminates and sequence number of each cable terminating at the equipment. Each cable shall be identified at both ends by tags bearing its assigned number. Tags shall be attached at each junction box and at the end of duct banks or tray systems.

4.5 Interface Cabinet (IFC)

IFC is for the termination of all field wirings and it is mounted / attached to RTU enclosure. It shall meet all the requirements of SEC Transmission Material Standard Specification 38-TMSS-03.

5.0 SUBSTATION SIGNAL POINT LIST

This section of the standard details the general requirements for substation Signal points at Control Panel and ECC (Energy Control Centre) via SCADA (Supervisory Control and Data Acquisition) intended to be used in the Transmission System of Saudi Electricity Company, Saudi Arabia. Final Signal point list including grouping of alarms shall be determined according to project requirements based on importance and shall be approved by SEC during detailed engineering stage.

This document specifies the trip and non-trip alarms, commands, status & monitoring of electrical parameters pertaining to different substation equipments. Depending upon the degree of essentiality, the alarms are organized on individual and group basis.

In addition, one contact of each initiating device shall be terminated on SCADA interface cabinet from where transmission of individual alarms (or group alarms) to ECC shall be possible.

To facilitate repair and maintenance of equipment in the Substation, the source of individual alarm or group alarms shall be displayed correspondingly in the Substation Annunciator Panel.

The substation alarms and their organization are covered in the following tables.

Table 27-3: Circuit Breaker 380/230/132/115/110/69/33 kV (As Applicable)

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	CIRCUIT BREAKER ABNORMAL.	1. Air pressure low stage-1 2. Air pressure high 3. Comp. Excessive running 4. Nitrogen volume Low 5. Oil pump / Air Compressor excessive running 6. Pump / Compressor excessive number of starts 7. Control supply supervision 8. Oil pressure low stage-1 9. Closing lock out 10. Closing spring faulty: a. Spring charger motor supply b. Supervision of MCB c. Spring not fully charged d. Compressor/pump motor supply 11. Test switch not in normal position 12. Breaker Lock out coil defective 13. BFR In/Out Switch 14. Closing Coil DC Supply Fail 15. BFR DC Fail 16. BFR Signal Fail 17. AC voltage monitoring relay fail for synch. check	****BREAKER ABNORMAL/ NORMAL OR ****ISOLATOR ABNORMAL/ NORMAL (As Applicable)	Non Trip
2	TRIP COIL-1 LOCK OUT	Trip Coil-1Lockout: a. SF6 low stage -2 lockout b. Low air stage -2 lockout c. Low oil stage-2 lockout d. Operating mechanism fail	**** LOCKOUT OPERATE/ NORMAL	Non Trip

Table 27-3: Circuit Breaker 380/230/132/115/110/69/33 kV (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
3	TRIP COIL-2 LOCK OUT	Trip Coil-2 Lockout: a. SF6 low stage-2 lockout b. Low air stage -2 lockout c. Low oil stage-2 lockout d. Operating mechanism fail	**** LOCKOUT OPERATE/ NORMAL	Non Trip
4	TRIP CIRCUIT-1 FAULTY.	Trip circuit supervision relay 27TC1	**** TRIP CIRCUIT FAULTY / NORMAL	Non Trip
5	TRIP CIRCUIT-2 FAULTY.	Trip circuit supervision relay 27TC2		
6	CBF PROTECTION FAULTY.	Circuit breaker fail protection supply supervision	**** CBF PROT. FAULTY/NORMAL	Non Trip
7	CBF PROTECTION OPERATED	Circuit breaker fail protection	**** CBF PROT. OPERATE/NORMAL	Trip
8	CB GAS PRESSURE LOW STAGE 1	Gas monitoring system	**** SF6 GAS ALARM/NORMAL	Non Trip
	CB GAS DENSITY HIGH			
9	REMOTE OFF.	Local / Remote switch of B.S. or B.C or Feeder Breaker or Transformer CB	**** REMOTE OFF / ON	Non Trip
10	CONTROL SUPPLY FAIL	Control supply supervision relay	**** AC/DC SP FAILED NORMAL	Non Trip
11	LCC DC SUPPLY FAIL	Supply supervision relay DC		
12	LCC AC SUPPLY FAIL	Supply supervision relay AC		Trip
13	CB PHASE DISCREPANCY	Phase / Pole discrepancy relay	**** PHNOTTOG OPERATE/ NORMAL	

Table 27-3: Circuit Breaker 380/230/132/115/110/69/33 kV (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
14	SYNCHRO-CHECK RELAY FAULTY	1. Synchrocheck Relay a. DC Failure b. Self Supervision 2. Synchronizing Switch in Off Position 3. Synchrocheck fuse failure relay operated	**** SYNC CHK FAULTY/ NORMAL	Non Trip
15	UNDER FREQUENCY	Under frequency protection relay	****U/F PROT. OPERATE/NORMAL	Trip
16	ALARM SUPPLY FAILURE	Alarm Indication Supply Failure	****ALARM SUPPLY FAILURE/NORMAL	Non Trip
17	ABTS FAILED	ABTS out of service a. AC Fail b. DC Fail c. PT Fuse supervision OPTD / Failure	****ABTS FAILED/NORMAL	Non Trip
18	CLOSE CIRCUIT FAULTY.	Closing Circuit supervision relay (27)	**** CLOSE CIRCUIT / FAULTY / NORMAL	Non Trip
19	CLOSE FAILURE PROTECTION OPERATED	Close Failure protection supervision. (For Series Compensation Breakers)	**** CLOSE FAILURE PROTECTION OPERATE / NORMAL	Non Trip
20	LINE CURRENT SUPERVISION OPERATED	Line Current Supervision (For Series Compensation Breakers)	**** LINE CURRENT SUPERVISION OPERATE / NORMAL	Non Trip
**** = Circuit Breaker / Isolator				

Table 27-4: Feeder 380kV OHL & U/G Cable (As Applicable)

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FEEDER PROTECTION FIRST MAIN FAULTY	1. Self diagnosis -System fail 2. Logic Fault 3. Pilot Faulty 4. VT Supervision 5. Pilot Supervision Supply 6. Protection Supply Supervision 7. Line protection un-stabilized 8. Trip Relay Coil Failure 9. Protection Trip (Switch) Isolated 10. Relay Set 1 (R1)/FPFM Blocked	**** PROT. FAULTY/ NORMAL	Non Trip
2	FEEDER PROTECTION SECOND MAIN FAULTY	Relay Set 2 (R2)/ FPSM Same as Feeder Protection First Main Faulty. (as applicable)		
3	BACKUP PROTECTION FAULTY	Backup protection relays		
4	PLC FAILED	PLC Equipment	**** PLC FAILED/ NORMAL	Non Trip
5	RELAY SET # 1 COMM. FAIL	Relay Set -1 Communication Fail a. PSE Fail b. PSE DC Fail c. Comm Channel Fail	R1/R2/OLS COMM FAILED/NORMAL	Non Trip
6	RELAY SET # 2 COMM. FAIL	Relay Set -2 Communication Fail a. PSE Fail b. PSE DC Fail c. Comm Channel Fail		

Table 27-4: Feeder 380kV OHL & U/G Cable (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
7	BKR FAIL COMM.-1 FAIL	Breaker Failure Communication -1 Fail a. PSE Fail b. PSE DC Fail c. Comm Channel Fail	BKR COMM 1/2 FAILED/NORMAL	Non Trip
8	BKR COMM.-2 FAIL	Breaker Failure Communication -2 Fail a. PSE Fail b. PSE DC Fail c. Comm Channel Fail		
9	AUTO RECLOSER FAULTY	1. Auto re-closer faulty 2. Auto re-closer out of service	**** AUTO REC. FAULTY/ NORMAL	Non Trip
10	AUTO RECLOSER BLOCKED	1. Auto re-closer selector switch in Off 2. Auto re-closer operation blocked	**** AUTO REC. BLOCK/ DEBLOCK	Non Trip
11	AUTO RECLOSER SUCCESSFUL	Auto recloser	**** AUTO REC. SUCCESSFUL	Non Trip
12	OIL PRESSURE LOW STAGE-1	Cable oil pressure low stage-1	**** OIL PR 1. LOW/ NORMAL	Non Trip
13	OIL PRESSURE LOW STAGE-2	Cable oil pressure low stage-2	**** OIL PR2. LOW/ NORMAL	Non Trip
14	INTER TRIP 1 FAULTY	1. Inter-trip channel faulty 2. Inter trip pilot faulty 3. Inter trip pilot supervision supply fail 4. Inter trip supply fail [As applicable]	**** INTER TRIP. FAULTY/NORMAL	Non Trip
15	INTER TRIP 2 FAULTY			
16	FEEDER END PROTECTION FAULTY	Feeder end Protection	**** FDR END FAULTY/ NORMAL	Non Trip
17	FEEDER END PROTECTION OPERATE	Feeder end Protection	**** FDR END OPERATE/ NORMAL	Trip
18	FERRO RESONANCE ALARM	Ferro resonance relay	**** FERRORES ALARM/ NORMAL	Non Trip

Table 27-4: Feeder 380kV OHL & U/G Cable (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
19	VT FAIL	1. VT secondary MCB / Fuse 2. Under voltage relay	****VT FAILED / NORMAL	Non Trip
20	OVER VOLTAGE PROTECTION OPERATED	Over voltage protection	****OVER VOLT. ALARM / NORMAL	Non Trip
21	OVER VOLTAGE PROTECTION FAULTY	Over voltage protection	****OVER VOLT. FAULTY / NORMAL	Non Trip
22	PHASE UNBALANCED	Phase unbalanced	****PH UNBAL. ALARM / NORMAL	Non Trip
23	SCADA ALARM SUPPLY OFF.	Selector switch	**** SCADA ALARM OFF/ON	Non Trip
24	DC 1 FAIL	Feeder DC -1 Failed	DC FAILED	Non Trip
25	DC 2 FAIL	Feeder DC -2Failed		
26	CABLE DIFF PROT OPERATED	Cable Differential Relay	**** CABLE DIFF. PROTECTION OPERATED/NORMAL	Trip
27	THERMAL O/L PROT. OPERATED	Cable thermal O/L Protection	**** CABLE THERMAL O/L PROTECTION OPERATED/NORMAL	Trip
**** = Breaker Number				

Table 27-5: Feeder 380kV Overhead Line

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FEEDER PROTECTION FIRST MAIN (RELAY SET 1) OPERATED	1. Zone 1 2. Zone 2 + Acc. Signal 3. DEF + Acc. signal	**** LINE PRO. OPERATE /NORMAL	Trip
2	FEEDER PROTECTION SECOND MAIN (RELAY SET 2) OPERATED	1. Zone 1 2. Zone 2 without blocking signal 3. DEF + Acc. signal		
3	BACKUP PROTECTION OPERATED	1. Close on to fault 2. Directional earth fault-backup + timer of FPFM 3. Zone 2 distance + timer of FPFM 4. Zone 3 distance + timer of FPFM 5. Directional earth fault backup + timer of FPSM 6. Zone 2 distance + timer of FPSM 7. Zone 3 distance + timer of FPSM 8. Tail End Cable Protection	****BKUP PRO. OPERATE /NORMAL	Trip
4	FIRST MAIN INTER TRIP RECEIVE	Inter trip receive relay 1	****INTRTRIP. RECEIVE /NORMAL	Trip
5	SECOND MAIN INTER TRIP RECEIVE	Inter trip receive relay 2		
**** = Breaker Number				

Table 27-6: Feeder 230/132/115/110/69 kV Overhead Line (As Applicable)

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FIRST MAIN PROTECTION OPERATED	1. Zone 1 2. DEF + Acc. signal	****LINE PRO. OPERATE /NORMAL	Trip
2	SECOND MAIN PROTECTION OPERATED	Directional earth fault + Acc signal		
3	BACKUP PROTECTION OPERATED	1. Close on to fault 2. Zone 2 plus backup timer 3. Zone 3plus backup timer 4. Over-current 5. Earth Fault 6. Directional earth fault plus backup timer	****BKUP PRO. OPERATE /NORMAL	Trip
4	FEEDER PROTECTION FIRST MAIN FAULTY	1. Self diagnosis -System fail 2. Logic Fault 3. Pilot Faulty 4. VT Supervision 5. Pilot Supervision Supply 6. Protection Supply Supervision 7. Line protection un-stabilized 8. Trip Relay Coil Failure 9. Protection Trip (Switch) Isolated	**** PROT. FAULTY/ NORMAL	Non Trip
5	FEEDER PROTECTION SECOND MAIN FAULTY	Same as Feeder Protection First Main Faulty. (as applicable)		
6	BACKUP PROTECTION FAULTY	Backup protection relays		
7	INTER TRIP RECEIVE	Inter trip receive relay	****INTR TRIP RECEIVE /NORMAL	Trip

Table 27-6: Feeder 230/132/115/110/69 kV Overhead Line (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
8	DC 1 FAIL	Feeder DC -1 Failed	DC FAILED (if required)	Non Trip
9	DC 2 FAIL	Feeder DC -2 Failed		
10	PROTECTION UNDER TEST	Feeder Protection Under Test mode		
**** = Breaker Number				

Table 27-7: Feeder 132/115/110/69 kV Underground Cable

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	MAIN PROTECTION OPERATED	Pilot wire protection	****LINE PRO OPERATE/NORMAL	Trip
2	BACKUP PROTECTION OPERATED	1. Backup over current 2. Backup earth fault		
3	INTER TRIP RECEIVE	Inter trip receive relay	****INTR TRIP RECEIVE/NORMAL	Trip
4	MAIN PROTECTION FAULTY	Pilot wire protection	****LINE PRO FAULTY/NORMAL	Non-Trip
5	BACKUP PROTECTION FAULTY	Backup Protection Relays		
6	CABLE DIFF PROT OPERATED	Cable Differential Relay	**** CABLE DIFF. PROTECTION OPERATED/NORMAL (if required)	Trip
7	THERMAL O/L PROT. OPERATED	Cable thermal O/L Protection	**** CABLE THERMAL O/L PROTECTION OPERATED/NORMAL	Trip
**** = Breaker Number				

Table 27-8: Feeder 33 kV Overhead Line

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Distance Protection Relay : 1. VT supply supervision 2. Logic Fault 3. Protection supply supervision	****FDR PROT FAULTY /NORMAL	Non-Trip
2	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Backup over current and earth fault protection relays supervision circuits		
3	INCOMER VT FAIL	Incomer VT Secondary MCB		
4	BUS VT FAIL	Bus VT Secondary MCB		
5	FLAGS ON RELAYS	Auto reclose relay	****AUTO REC BLOCK /DEBLOCK	Non-Trip
6	FLAGS ON RELAYS	Trip circuit supervision	***TRIP CKT FAULTY /NORMAL	Non-Trip
7	FLAGS ON RELAYS	Distance Protection Relay	**** FDR PROT. OPERATE/NORMAL	Trip
8	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Directional earth fault relay		
9	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Backup O/C & E/F protection relay		
10	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Sensitive earth fault protection relay		
11	FLAGS ON RELAYS	Auto reclose relay	****AUTO REC FAULTY /NORMAL	Non-Trip
**** = Breaker Number				

Table 27-9: Feeder 33 kV Underground Cable

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FLAGS ON RELAYS	Pilot wire supervision relay	**** FDR PROT. FAULTY /NORMAL	Non-Trip
2	FLAGS ON RELAYS	Protection supply supervision relay for pilot wire and backup protection		
3	FLAGS ON RELAYS	Pilot wire protection relay	**** FDR PROT. OPERATE/NORMAL	Trip
4	FLAGS ON RELAYS	Backup over current & earth fault protection relay		
**** = Breaker Number				

Table 27-10: 13.8 kV Feeders/Incomers/Tie-Breakers Grouped Alarms (As applicable)

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Auto re-closer relay (OHL)	13.8 kV AUTO REC BLOCK/DEBLOCK	Non-Trip
2	FLAGS ON RELAYS	Trip circuit supervision relay-1 Trip circuit supervision relay-2	13.8 kV PROT FAULTY /NORMAL	Non-Trip
3	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Auto re-closer relay (OHL)		
4	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Line protection relay		
5	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Line protection relay	13.8 kV PROT OPERATE /NORMAL	Trip
6	CONTROL SUPPLY FAIL	Control supply supervision relay	13.8kV AC/DC SUPPLY FAILED /NORMAL	Non-Trip
7	DC SUPPLY FAILURE	Supply supervision relay DC		
8	AC SUPPLY FAILURE	Supply supervision relay AC		
9	CLOSING CIRCUIT DC SUPPLY FAIL	Closing Circuit Supervision relay (27)	**** CLOSE CIRCUIT / FAULTY / NORMAL	Non Trip
10	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	SF6 gas low	13.8 kV SF6 GAS ALARM /NORMAL	Non-Trip
11	TRIP COIL LOCK OUT	Trip Coil Lockout : 1.SF6 lockout 2.Low air lockout 3.Low oil lockout	13.8kV LOCKOUT OPERATE/NORMAL	Non-Trip

Table 27-10: Feeder 13.8 kV Feeders/Incomers/Tie-Breakers Grouped Alarms *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
12	FLAG ON AUX.RELAY / DISPLAY ON NUMERICAL RELAY	Pilot Wire Protection	13.8kV P.W PROT OPERATE/NORMAL	Trip
13	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Remote end push button local end auxiliary relay	13.8kV REMOTE TRIP RECEIVE/NORMAL	Trip
14	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Alarm Indication Supply Failure	13.8kV ALARM SUPPLY FAILED /NORMAL	Non-Trip
15	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	ABTS Out of Service a. AC Fail b. DC Fail c. PT Fuse supervision OPTD / Failure	13.8 kV ABTS FAILED/NORMAL	Non-Trip
16	FLAG ON SYNCHRONISING RELAY	Synchronizing Failure	13.8 kV SYNCH FAILED/NORMAL	Non-Trip
17	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	13.8 kV Bus/Incomer VT Failure	13.8 kV VT FAILED/NORMAL	Non-Trip
18	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Local/Remote Switch Left to Local	"CB #" Control Local	Non-Trip
Note for Item # 12 & 13 : In case of 13.8 kV consumer feeder with pilot wire protection, these two additional alarms to SCADA shall be required				

Table 27- 11: 380/230;132;115;110 kV & 230/69 kV Transformers (As Applicable)

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
380 kV CONTROL PANEL				
1	T # FIRST MAIN PROTECTION FAULTY	1.First main protection CT circuit supervision relay 2. DC Fail 3. Lock out Relay Fail	T# PROTECTION FAULTY /NORMAL	Non-Trip
2	T # SECOND MAIN PROTECTION FAULTY	1. Second main protection CT circuit supervision relay 2. DC Fail 3. Lock out Relay Fail		
3	T # BUCHHOLZ STAGE -I	Main Transformer Buchholz stage-I	T# BUCHHOLZ ST1 ALARM /NORMAL	Non-Trip
4	T # AUX/EARTHING TRAFO BUCHHOLZ STAGE-1	1.Aux Transformer Buchholz stage-I 2.Earthing Trafo buchholz stage-I		
5	T# WINDING TEMPERATURE HIGH STAGE -I	1.HV winding temperature high stage-I. 2.LV winding temperature high stage-I. 3.Tertiary wdg temp High stage-1	T# WDG TEMP ST1 ALARM/NORMAL	Non-Trip
6	T # AUX TRAFO WINDING TEMPERATURE HIGH STAGE -I	Auxiliary transformer winding temperature high stage -I.		
7	T # OIL TEMPERATURE HIGH STAGE-1	Main Transformer Oil Temperature Thermometer Stage-1	T # OIL TEMP ST1 ALARM /NORMAL	Non-Trip
8	T # AUX TRAFO OIL TEMPERATURE HIGH STAGE-1	Aux. Transformer Oil Thermometer- Stage -1		
9	T # OIL TEMPERATURE HIGH STAGE-2	Main Transformer Oil Temperature Thermometer Stage-2	T # OIL TEMP ST2 ALARM /NORMAL	Non-Trip
10	T # AUX TRAFO OIL TEMPERATURE HIGH STAGE-2	Aux. Transformer Oil Thermometer- Stage -2		
11	T# OIL LEVEL LOW ALARM	1.Main transformer oil level low. 2.Tap changer oil level low.	T # TRAFO OIL LVL LOW /NORMAL	Non-Trip
12	T# AUX TRAFO OIL LEVEL LOW ALARM	Auxiliary transformer oil level low.		

Table 27- 11: 380/230;132;115;110 kV & 230/69 kV Transformers (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
13	T # PRESSURE RELIEF OPERATED	1. Main transformer pressure relief trip 2. Tap changer pressure relief trip	T # TRAFO PRESS RELIEF TRIP/NORMAL	Trip
14	T # CABLE OIL PRESSURE LOW	1. HV Cable Oil Pressure Low Stage-1 2. HV Cable Oil Pressure Low Stage-2	T # CABLE OIL PRESS LOW /NORMAL	Non-Trip
15	COOLING SYSTEM FAULTY	1. Fan MCB Trip 2. Oil pump MCB Trip 3. Zero oil flow 4. Cooler AC/DC supply Fail 5. AC System Voltage Fail	T # COOLING SYSTEM FAULTY /NORMAL	Non-Trip
16	FIRE PROTECTION FAULTY	1. Heat detector open circuited or removed. 2. Battery/AC supply disconnected	T # FIRE PROT FAULTY /NORMAL	Non-Trip
17	FIRE PROTECTION OPERATED	1. Operation of pressure switch in the respective transformer deluge valve due to transformer water spray discharge 2. Operation transformer water spray system manual release station 3. Transformer heat detector operated	T# FIRE PROT OPERATE /NORMAL	Non-Trip
18	T# TAP CHANGER FAULTY	1. Tap Changer Drive Failure 2. OLTC Parallel Operation disturbed 3. OLTC Raise/Lower Limit Reached 4. OLTC Motor Circuit Tripped 5. Loss of OLTC Motor Supply 6. OLTC out of step 7. AVR Blocked (O/V, O/C, U/V) 8. RTCC Control Supply Fail	T# TAP CHANGER DRIVE FAILED/NORMAL	Non-Trip
19	T# TRAFO TRIP DISABLED	Transformer Trip Disabled	T# TRANSFORMER TRIP DISABLED	Non-Trip

Table 27- 11: 380/230;132;115;110 kV & 230/69 kV Transformers (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
20	T # FIRST MAIN PROTECTION OPERATED	1. First circulating current protection 2. Tertiary restricted earth fault 3. Tertiary instantaneous over current 4. HV instantaneous over current 5. HV over current stage-1 6. HV over current stage-2 7. Main transformer buchholz stage-2 8. Aux transformer buchholz stage -2 9. OLTC Surge Relay Trip	T # PROTECTION OPERATED /NORMAL	Trip
21	T # SECOND MAIN PROTECTION OPERATED	1. Second circulating current protection 2. Tertairy over current protection 3. Tertiary earth fault protection 4. Power transformer LV over current and earth fault protection 5. 220 V restricted earth fault 6. Winding temperature stage-2		
22	T # BACKUP PROTECTION OPERATED	Backup over current and earth fault protection		
23	WINDING TEMPERATURE HIGH STAGE-2	1. Main T/F HV winding temperature high stage-2 2. Main T/F LV winding temperature high stage-2 3. Main T/F tertiary winding temperature stage-2 4. Earthing / Aux Transformer winding Temperature Stage- 2	T# WDG TEMP ST2 OPERATE /NORMAL	Trip

Table 27- 11: 380/230;132;115;110 kV & 230/69 kV Transformers (As Applicable) *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
132/115/110 kV CONTROL PANEL				
24	T # FIRST PROTECTION SUPPLY SUPERVISION	First protection supply supervision relay	T # PROTECTION FAULTY /NORMAL	Non-Trip
25	T # SECOND PROTECTION SUPPLY SUPERVISION	Second protection supply supervision relay		
26	T # BACKUP PROTECTION FAULTY	Backup over current and earth fault protection.		
27	T # FIRST MAIN PROTECTION LV TRIP-1 OPERATED	LV Trip Relay	T # PROTECTION OPERATE/NORMAL	Trip
28	T # FIRST MAIN PROTECTION LV TRIP-2 OPERATED	LV Trip Relay		
29	FLAG RELAY	VT supply failed	T # *VT FAILED/NORMAL	Non-Trip
# = Transformer Number ; * = Transformer LV Breaker Number				

Table 27-12: 132;115;110/33 kV Or 132;115; 110/13.8kV Transformers

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	TRANSFORMER OIL LEVEL LOW	1. Main transformer oil level low 2. Tap changer oil level low 3. Auxiliary transformer oil level low	T # OIL LEVEL LOW / NORMAL	Non-Trip
2	MAIN TRANSFORMER OIL TEMP HIGH	Main Transformer Oil Thermometer	T# OIL TEMP HIGH/NORMAL	
	OIL TEMPERATURE HIGH STAGE 1			
	OIL TEMPERATURE HIGH STAGE 2			
3	AUX TRANSFORMER OIL TEMP HIGH	Aux Transformer Oil Thermometer		
	OIL TEMPERATURE HIGH STAGE 1			
	OIL TEMPERATURE HIGH STAGE 2			
4	TRANSFORMER FAULTY	1. Main Pressure Relief Operated 2. Tap changer pressure relief Operated	T # PRESS RELIEF OPERATED	Operated
5	COOLER FAIL	1. Fan MCB 2. Oil pump MCB 3. Zero oil flow 4. Cooler AC/DC supply supervision	T # COOLING SYSTEM FAULTY /NORMAL	Non-Trip
6	PROTECTION FAULTY	Main and backup protection supply supervision	T # PROTECTION FAULTY/NORMAL	Non-Trip
7	BUCHHOLZ STAGE-1	1. Main transformer buchholz stage -1	T # BUCHHOLZ ST1 ALARM/NORMAL	Non-Trip
8	WINDING TEMPERATURE HIGH STAGE-1	1. Main transformer LV winding temperature high stage-I 2. Main transformer HV winding temperature high stage-I 3. Main transformer tertiary winding temperature high stage-I	T # WDG TEMP ST1 ALARM/NORMAL	Non-Trip
		4. Auxiliary transformer winding temperature high stage -I		

Table 27-12: 132;115;110/33 kV Or 132;115; 110/13.8kV Transformers *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
9	TRANSFORMER HV REF. OPERATED	HV restricted earth fault operated	T # PROTECTION OPERATE/NORMAL	Trip
10	TRANSFORMER LV REF. OPERATED	LV restricted earth fault operated		
11	TRANSFORMER DIFFERENTIAL PROT. OPERATED	Biased differential protection operated		
12	TRANSFORMER BUCHHOLZ STAGE-2 OPERATED	1. Main transformer buchholz stage-2 2. Tap changer buchholz protection	T # PROTECTION OPERATE/NORMAL	Trip
13	TRANSFORMER STANDBY EARTH FAULT PROT. OPERATED	Stand by earth fault protection		
14	TRANSFORMER HV O/C & E/F PROT. OPERATED	HV Over current and earth fault protection		
15	TRANSFORMER LV O/C & E/F PROT OPERATED	LV Over current and earth fault protection		
16	WINDING TEMPERATURE HIGH STAGE-2	1. Main transformer HV winding temperature high stage-2 2. Main transformer LV winding temperature high stage-2	T # WDG TEMP ST2 OPERATE/NORMAL	Trip
17	FLAG RELAY	VT supply failed	T # * VT FAILED/NORMAL	Non-Trip
18	FIRE PROTECTION OPERATED	1. Transformer heat detector operated 2. Operation pressure switch in the respective deluge valve to transformer water spray 3. Operation of transformer water spray manual release station 4. Transformer fire protection operated. (SERGI)	T # FIRE PROT OPERATE/NORMAL	Non-Trip
19	FIRE PROTECTION FAULTY	1. Heat detector open circuited or removed 2. Transformer fire protection SERGI 3. Battery/AC supply disconnected	T # FIRE PROT FAULTY/NORMAL	Non-Trip

Table 27-12: 132;115;110/33 kV Or 132;115; 110/13.8kV Transformers *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
20	T# TAP CHANGER FAULTY	1. Tap Changer Drive Failure 2. OLTC Parallel Operation disturbed 3. OLTC Raise/Lower Limit Reached 4. OLTC Motor Circuit Tripped 5. Loss of OLTC Motor Supply 6. OLTC out of step 7. AVR Blocked (O/V, O/C, U/V) 8. RTCC Control Supply Fail	T# TAP CHANGER DRIVE FAILED/NORMAL	Non-Trip
21	T# TRAFO TRIP DISABLED	Transformer Trip Disabled	T# TRANSFORMER TRIP DISABLED	Non-Trip
22	UNDER FREQUENCY PROTECTION FAULTY	Under frequency protection relay	T # UNDER FREQ RLY FAULTY/NORMAL	Non-Trip
23	UNDER FREQUENCY PROT OPERATED	Under frequency protection relay	T# UNDER FREQ RLY OPERATE/NORMAL	Trip
24	AUXILIARY TRANSFORMER PROTECTION OPERATED	LV main circuit breaker OFF	AUX T # PROTECTION OPERATE/NORMAL	Trip
25	AUXILIARY TRANSFORMER PROTECTION FAULTY	Auxiliary transformer HV fuses blown out	AUX T # PROTECTION FAULTY/NORMAL	Non-Trip
26	UNDER VOLTAGE PROTECTION FAULTY	Under voltage relay	T # UNDER VOLT RLY FAULTY/NORMAL	Non-Trip
# = Transformer Number; * = Transformer LV Breaker Number				

Table 27-13: 33/13.8 kV Transformers

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	1.Main transformer buchholz stage -1.	T # BUCHHOLZ ALARM/NORMAL	Non-Trip
2	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	1.Transformer oil level low. 2.Tap changer oil level low. 3.Oil temperature high stage -1. 4.Oil temperature high stage-2. 5.Pressure relief device operated.	T # TRANSFORMER ALARM/NORMAL	Non-Trip
3	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	1.Control supply fail. 2.Control circuit fault. 3.Fan supply MCB off. 4.Cooler AC/DC supply supervision.		
4	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Transformer winding temperature high stage - 1.	T # WDG. TEMP. ST1 ALARM/NORMAL	Non-Trip
5	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	HV protection supply supervision.	T # PROTECTION FAULTY/NORMAL	Non-Trip
6	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	LV protection supply supervision.		
7	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Transformer winding temperature high stage - 2.	T # PROTECTION OPERATE/NORMAL	Trip
8	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Buchholz protection.		

Table 27-13: 33/13.8 kV Transformers *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
9	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Transformer differential protection (87T)	T # PROTECTION OPERATE/NORMAL	Trip
10	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Transformer restricted earth fault REF (64S)		
11	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	HV Over current and earth fault protection		
12	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Stand by earth fault protection		
13	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Transformer LV over current & earth fault protection		
# = Transformer Number				

Table 27-14: Reactors

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	REACTOR FIRST PROTECTION MAIN FAULTY	Reactor circulating current protection supervision	**** PROT FAULTY/NORMAL	Non-Trip
2	REACTOR SECOND PROTECTION MAIN FAULTY	Reactor circulating current protection supervision		
3	REACTOR CONNECTION PROTECTION 1 FAULTY	Reactor connection circulating current protection supervision relay		
4	REACTOR CONNECTION PROTECTION 2 FAULTY	Reactor connection circulating current protection supervision relay		
5	REACTOR BACKUP PROTECTION FAULTY	Reactor backup over current and earth fault protection supervision		
6	REACTOR VOLTAGE UNBALANCE DETECTED	Reactor Voltage Unbalance Supervision		
7	REACTOR WINDING TEMPERATURE STAGE -1 OPERATED	Reactor winding temperature high stage-1	**** WDG TEMP. ST1 ALARM/NORMAL	Non-Trip
8	REACTOR WINDING TEMPERATURE STAGE -2 OPERATED	Reactor winding temperature high stage-2	**** WDG TEMP. ST2 ALARM/NORMAL	Trip
9	REACTOR BUCHHOLZ STAGE-1	Reactor buchholz stage -1	****BUCHHOLZ ST1 ALARM/NORMAL	Non-Trip
10	REACTOR FAULTY	1. Oil level low 2. Oil level high 3. Pressure relief 4. Oil temperature high Stage -1 5. Cooling Fail	**** REACTOR ALARM/NORMAL	Non-Trip

Table 27-14: Reactors *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
11	REACTOR FIRE PROTECTION OPERATED	1. Operation of pressure switch in the respective transformer deluge valve due to reactor water spray discharge 2. Operation reactor water spray system manual release station 3. Reactor heat detector operated	**** FIRE PROT. OPERATE/NORMAL	Non-Trip
12	REACTOR OIL TEMPERATURE VERY HIGH	Oil Temperature High Stage -2	**** REACTOR OIL TEMP VERY HIGH / NORMAL	Non-Trip
13	REACTOR FIRE PROTECTION FAULTY	1. Heat detector open circuited or removed 2. Battery/AC supply disconnected	**** FIRE PROT. FAULTY/NORMAL	Non-Trip
14	REACTOR CONTROL AC/DC SUPPLY FAIL	Control supply supervision relay	**** AC/DC SP. FAILED/NORMAL	Non-Trip
15	AC SUPPLY TRIP	AC supervision relay		
16	DC SUPPLY TRIP	DC supervision relay		
17	REACTOR FIRST PROTECTION MAIN OPERATED	Reactor circulating current protection-1	**** PROT OPERATE/NORMAL	Trip
18	REACTOR SECOND MAIN PROTECTION OPERATED	Reactor circulating current protection-2		
19	REACTOR CONNECTION PROTECTION-1 OPERATED	Reactor connection circulating current protection-1		
20	REACTOR CONNECTION PROTECTION -2 OPERATED	Reactor connection circulating current protection-2		

Table 27-14: Reactors *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
21	REACTOR BACKUP PROTECTION OPERATED	Reactor backup over current and earth fault protection	**** PROT OPERATE/NORMAL	Trip
22	REACTOR BUCHHOLZ STAGE-2	Reactor buchholz stage -2		
**** = Reactor Number				

Table 27-15: Bus with Busbar Protection 380/230/132/115/110/69 kV (As Applicable)

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	BUS BAR ZONE1A PROTECTION FAULTY	1. Bus bar protection out of service 2. Bus bar protection faulty 3. CT bus wire supervision 4. Interpose relay faulty 5. Supply supervision 6. Zones interconnected 7. CT bus wire shorting operated 8. Trip lockout relay faulty	BUS 1A BUSBAR PROT FAULTY/NORMAL	Non-Trip
2	BUS BAR ZONE1B PROTECTION FAULTY	1. Bus bar protection out of service 2. Bus bar protection faulty 3. CT bus wire supervision 4. Interpose relay faulty 5. Supply supervision 6. Zones interconnected 7. CT bus wire shorting operated 8. Trip lockout relay faulty	BUS 1B BUSBAR PROT FAULTY/NORMAL	Non-Trip
3	BUS BAR ZONE 2A PROTECTION FAULTY	1. Bus bar protection out of service 2. Bus bar protection faulty 3. CT bus wire supervision 4. Interpose relay faulty 5. Supply supervision 6. Zones interconnected 7. CT bus wire shorting operated 8. Trip lockout relay faulty	BUS 2A BUSBAR PROT FAULTY/NORMAL	Non-Trip

Table 27-15: Bus with Busbar Protection 380/230/132/115/110/69 kV *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
4	BUS BAR ZONE 2B PROTECTION FAULTY	1. Bus bar protection out of service 2. Bus bar protection faulty 3. CT bus wire supervision 4. Interpose relay faulty 5. Supply supervision 6. Zones interconnected 7. CT bus wire shorting operated 8. Trip lockout relay faulty	BUS 2B BUSBAR PROT FAULTY/NORMAL	Non-Trip
5	BUS BAR PROTECTION CHECK ZONE FAULTY	1. Bus bar check zone protection supervision relay 2. Check zone prot "out" 3. CT supervision 4. Trip lockout relay faulty	BB CHK.PROT. FAULTY/NORMAL	Non-Trip
6	BUS BAR ZONE1A PROTECTION OPERATED	1. Busbar zone 1A protection lockout (86B1) 2. Busbar zone 1A protection lockout (86B2)	BUS 1A BUSBAR PROT OPERATE/NORMAL	Trip
7	BUS BAR ZONE1B PROTECTION OPERATED	1. Busbar zone 1B protection lockout (86B1) 2. Busbar zone 1B protection lockout (86B2)	BUS 1B BUSBAR PROT OPERATE/NORMAL	Trip
8	BUS BAR ZONE 2A PROTECTION OPERATED	1. Busbar zone 2A protection lockout (86B1) 2. Busbar zone 2A protection lockout (86B2)	BUS 2A BUSBAR PROT OPERATE/NORMAL	Trip
9	BUS BAR ZONE 2B PROTECTION OPERATED	1. Busbar zone 2B protection lockout (86B1) 2. Busbar zone 2B protection lockout (86B2)	BUS 2B BUSBAR PROT OPERATE/NORMAL	Trip
10	BUS CHECK ZONE PROTECTION OPERATED	Busbar check zone 1A/2A/1B/2B protection lockout	BUSBAR CHECK ZONE PROT OPERATE/NORMAL	Trip

Table 27-15: Bus with Busbar Protection 380/230/132/115/110/69 kV *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
11	BUS COUPLER / TIE OVER CURRENT PROTECTION OPERATED	Bus coupler over current protection	*** O/C PROT. OPERATE/NORMAL	Trip
12	BUS COUPLER / TIE OVER CURRENT PROTECTION FAULTY	1. Protection supply supervision 2. Relay internal fault	*** O/C PROT. FAULTY/NORMAL	Non-Trip
13	BUS SECTION OVER CURRENT PROTECTION OPERATED	Bus section over current protection	*** O/C PROT. OPERATE/NORMAL	Trip
14	BUS SECTION OVER CURRENT PROTECTION FAULTY	1. Protection supply supervision 2. Relay internal fault	*** O/C PROT. FAULTY/NORMAL	Non-Trip
15	BUS VT FAILED	VT secondary MCB OFF	BB VT. FAILED/NORMAL	Non-Trip
16	BUSBAR GAS PRESSURE ALARM	1. Busbar SF6 gas pressure low stage-1 2. Busbar has pressure high	BUS_ *GIS SF6 ALARM/NORMAL	Non-Trip
17	BUSBAR GAS PRESSURE LOW STAGE-2	Busbar gas pressure low stage -2	BUS_ *GIS SF6 ST2 ALARM/NORMAL	Non-Trip
18	BUSBAR PROTECTION TRIP SUPPLY -1 SUPERVISION	1. Relay flag 2. Actuator position	BB PROT TRIP SUPPLY-1 FAULTY/NORMAL	Non-Trip
19	BUSBAR PROTECTION TRIP SUPPLY -2 SUPERVISION	1. Relay flag 2. Actuator position	BB PROT TRIP SUPPLY-2 FAULTY/NORMAL	Non-Trip
20	BUSBAR LOCKOUT RESET SUPPLY FAIL	1. Relay flag 2. Actuator position	BB PROT RESET SUPPLY FAIL/NORMAL	Non-Trip

Table 27-15: Bus with Busbar Protection 380/230/132/115/110/69 kV *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
MESH CORNER TYPE SUBSTATIONS				
21	MESH CORNER PROTECTION FAULTY	1. Bus bar protection out of service 2. Bus bar protection faulty 3. CT bus wire supervision. 4. Interpose relay faulty 5. Supply supervision 6. CT bus wire shorting operated one for each mesh corner	MESH COR*PROT FAULTY/NORMAL	Non-Trip
22	MESH CORNER PROTECTION OPERATED	Mesh corner protection one for each mesh corner	MESH COR*PROT OPERATE/NORMAL	Trip
33 kV SINGLE BUSBAR TYPE SUBSTATIONS				
23	BUSBAR * PROTECTION FAULTY	1. Bus bar protection out of service 2. Bus bar protection faulty 3. CT bus wire supervision 4. Interpose relay faulty 5. Supply supervision 6. CT bus wire shorting operated one for each mesh zone	BUSBAR *PROT FAULTY/NORMAL	Non-Trip
24	BUSBAR * PROT. OPERATED	Bus bar protection one for each zone.	BUSBAR *PROT OPERATE/NORMAL	Trip
25	BUS VT FAILED	VT secondary MCB OFF	BB VT. FAILED/NORMAL	Non-Trip
13.8 kV SINGLE BUSBAR				
26	13.8 BUSBAR STUCK BREAKER PROTECTION FAULTY	1. Stuck breaker protection supply supervision 2. Protection relay failure	BUS*STUCK. CB. PROT FAULTY/NORMAL	Non-Trip
27	13.8 BUSBAR STUCK BREAKER PROT OPERATED	Stuck breaker protection	BUS*STUCK. CB. PROT OPERATE/NORMAL	Trip
* = Busbar Section				

Table 27-16: Substation General Alarms

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
380 / 230;132;115;110 kV , 230 / 69 kV AND 132;115;110 / 33 kV ; 13.8 kV SUBSTATIONS				
1	EMERGENCY DIESEL FAULT	Diesel generator fault	DIESEL GEN. FAULTY/NORMAL	Non-Trip
2	125 VOLT BATTERY # 1EARTH FAULT	125 V battery # 1 earth fault relay	125 V DC GROUND ALARM/NORMAL	Non-Trip
3	125 VOLT BATTERY # 2 EARTH FAULT	126 V battery # 2 earth fault relay		
4	125 VOLT BATTERY # 1 FAULTY	1. Battery volt low 2. Battery volt high 3. Battery high impedance 4. Battery charger failure 5. Battery charger AC supply failure 6. Charger temperature high 7. Charger over voltage	125V BATTERIES FAULTY/NORMAL	Non-Trip
5	125 VOLT BATTERY # 2 FAULTY	1. Battery volt low 2. Battery volt high 3. Battery high impedance 4. Battery charger failure 5. Battery charger AC supply failure 6. Charger temperature high 7. Charger over voltage		
6	48 V BATTERY EARTH FAULT	48 V battery earth fault relay	48 V DC GROUND ALARM/NORMAL	Non-Trip

Table 27-16: Substation General Alarms *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
7	48 VOLT BATTERY FAULTY	1. Battery volt low 2. Battery volt high 3. Battery high impedance 4. Battery charger failure 5. Battery charger AC supply failure 6. Charger temperature high 7. Charger over voltage	48 V BATTERY FAULTY/NORMAL	Non-Trip
8	EXHAUST FAN FAIL	Battery Room Exhaust Fan Failed	EXHAUST FAN FAIL/NORMAL	Non-Trip
9	SMOKE DETECTOR OPERATED	1. Smoke detector operated 2. Manual call point operated	SMOKE DETECTOR OPERATE/NORMAL	Non-Trip
10	ROOM TEMPERATURE HIGH	Room temperature high	ROOM TEMPERATURE HIGH/NORMAL	Non-Trip
11	RTU POWER SUPPLY FAIL	RTU Power Supply a. AC failed. b. DC failed	RTU POWER SUPPLY NORMAL/FAILED	Non-Trip
12	----------	Entrance alarm-1	ENTRANCE ALARM/NORMAL	Non-Trip
13	----------	Entrance alarm-2	INTRUDER ALARM/NORMAL	Non-Trip
14	ISOLATOR SWITCHING INCOMPLETE	Isolator stuck in the middle position during operation	SWITCHING INCOMPLETE ALARM/NORMAL	Non-Trip
15	EARTH SWITCH SWITCHING INCOMPLETE	Earth switch stuck in the middle position during operation		
16	PLC CHARGER FAULTY	1. PLC 48V battery 2. PLC battery charger 3. PLC battery charger AC supply failed	PLC CHARGER FAULTY/NORMAL	Non-Trip

Table 27-16: Substation General Alarms *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
17	FIRE SYSTEM FAULTY	1. Diesel pump faulty 2. Electric pump faulty 3. Fire pump faulty 4. Fire system compressor faulty 5. Fire alarm / Extinguishing (FM200/CO2/Halon/System faulty) 6. FM200/CO2/Halon system in manual mode	FIRE SYSTEM FAULTY/NORMAL	Non-Trip
18	OLS CHARGER FAULTY.	1. OLS battery voltage low 2. OLS battery voltage high 3. OLS battery high impedance	OLS CHARGER FAULTY/NORMAL	Non-Trip
19	S/S AUXILIARY SUPPLY FAILS AC MCB OFF.	1. Substation auxiliary supply fail 2. AC distribution panel MCB	S/S AUXILIARY SUPPLY FAILED/NORMAL	Non-Trip
20	MAIN DC SUPPLY FAILED	380, 132, 33 &13.8 kV control board main battery DC failure	MAIN DC SUPPLY FAILED /NORMAL	Non-Trip
21	DC MCB OFF.	DC Distribution panel MCB		
22	COMPRESSOR FAULTY	1. Compressor excessive running 2. Compressor CB 3. Air pressure low	COMPRESSOR FAULTY/NORMAL	Non-Trip
23	NEUTRAL CURRENT ALARM OPERATED	Neutral current alarm operated	NEUTRAL CURRENT OPERATE/NORMAL	Non-Trip
24	FIRE SYSTEM OPERATED	1. Main electric pump operate 2. Diesel engine driven pump operate 3. Fire system compressor operate 4. Operation of Halon /CO2/FM200/pressure switch on system header 5. FM200/Halon/CO2 /manual release unit operated	FIRE SYSTEM OPERATE/NORMAL	Non-Trip

Table 27-16: Substation General Alarms *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
25	OLS # FAILED	OLS # system	OLS # FAILED/NORMAL	Non-Trip
26	OLS SYSTEM FAULTY	OLS # system faulty		
27	ALARM SYSTEM FAULTY	Alarm system supply failed	ALARM SYSTEM FAULTY/NORMAL	Non-Trip
28	HVAC SYSTEM MAIN FAULTY	1. Air conditioning system 2. Ventilation	HVAC SYSTEM MAIN FAULTY / NORMAL	Non-Trip
	HVAC SYSTEM BACKUP FAULTY	1. Air Conditioning system 2. Ventilation	HVAC SYSTEM BACK UP FAULTY / NORMAL	Non-Trip
29	AUTO RESET SUPPLY FAIL	Auto reset DC supply for lockout relays	AUTO RESET FAULTY/NORMAL	Non-Trip
30	RTU CHARGER FAULTY	1. RTU 48 V battery 2. RTU battery charger 3. RTU battery charger AC supply failed	RTU CHARGER FAULTY/NORMAL	Non-Trip
31	SYSTEM SYNCHRONIZER UNIT FAULTY	System synchronizer	SYS.SYNC. FAULTY/NORMAL	Non-Trip
32	U/F RELAY FAIL	Under frequency protection relay	****U/F PROT. FAULTY /NORMAL	Non -Trip
AUTOMATIC SWITCHING				
33	AUTOMATIC SWITCHING FAILED	Instruction control time out	AUTO SWITCHING. FAILED/NORMAL	Non-Trip
34	AUTOMATIC SWITCHING EQUIPMENT FAULTY	1. Logic Faulty 2. Processor watchdogs 3. Supply supervision 4. Input discrepancy 5. Automatic switching .Equipment out of service	AUTO SWITCHING EQUIP FAULTY/NORMAL	Non-Trip

Table 27-16: Substation General Alarms *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
35	TRIP ON PARALLEL	Trip on parallel selected	TRIP ON PARALLEL	Non-Trip
36	AUTO SW.EQ. REMOTE	Auto Sw. Eq. selected to Remote	AUTO SW EQ. REMOTE ON/OFF	Non-Trip
CHANGEOVER EQUIPMENT ALARMS				
37	ACSE REMOTE	ACSE selected to Remote	ACSE REMOTE ON/OFF	Non-Trip
38	AUTO CHANGEOVER FAILED	Instruction control time out	AUTO CHANGEOVER. FAILED/NORMAL	Non-Trip
39	AUTO CHANGEOVER FAULTY	Auto changeover system equipment supply supervision	AUTO CHANGEOVER. FAULTY/NORMAL	Non-Trip
FAULT RECORDER ALARMS				
40	FAULT RECORDER OPERATED	Fault recorder operated	FAULT RECORDER OPERATE/NORMAL	Non-Trip
41	FAULT RECORDER FAULTY	1. Logic Faulty 2. Memory full 3. Out of paper 4. Paper low 5. Processor watchdogs 6. Printer faulty 7. Supply supervision 8. Out of service	FAULT RECORDER FAULTY/NORMAL	Non-Trip

Table 27-16: Substation General Alarms *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
42	ROOM TEMPERATURE HIGH	Room temperature high	ROOM TEMPERATURE HIGH/NORMAL	Non-Trip
43	RTU DC LOW	1. RTU DC low 2. Command power failed	RTU DC FAILED/NORMAL	Non-Trip
44	------	Entrance - 1	ENTRANCE ALARM/NORMAL	Non-Trip
45	------	Entrance - 2	INTRUDER ALARM/NORMAL	Non-Trip
46	NEUTRAL CURRENT ALARM OPERATED	Neutral current alarm relay	NEUTRAL CURRENT OPERATE/NORMAL	Non-Trip
47	COMMUNICATION FAIL	Major Communication Failure	MAJ COMM FAIL/NORMAL	Non-Trip

Table 27-17: Automatic Voltage Control

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	TAP CHANGER INCOMPLETE	Tap changer incomplete	**TAP INCOMPLETE ALARM/NORMAL	Non-Trip
2	VOLTAGE CONTROL FAIL	Automatic Voltage Control Equipment OR Tap Changer	**AVC/TAP CHANGER FAULTY/NORMAL	Non-Trip
3	TAP CHANGER MOTOR FAIL			
4	AVC SUPPLY FAULTY			
5	TAP CHANGER OUT OF STEP			
6	FLAG RELAY	Tap changer in progress	**TAP CHANGER IN-PROGRESS / FINISHED	Non-Trip
** = Transformer Number				

Table 27-18: Thyristor Controlled Reactor Alarms

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	DISPLAY ON REACTOR CONTROL PANEL	Cooling System Failure 1. Cooling Fan Fail 2. Cooling Water Temp. High 3. Cooling Pump Fail 4. Cooling Supply Fail 5. Cooling Water Level Low 6. Cooling Water Conductivity High 7. Cooling System Trip 8. TCR Water Heater fail.	TCR #COOLING SYSTEM FAILED/NORMAL	Trip
2	DISPLAY ON REACTOR CONTROL PANEL	Buchholz Stage 2	TCR #BUCHHOLZ ST 2 TRIP / NORMAL	Trip
3	DISPLAY ON REACTOR CONTROL PANEL	TCR Controller Fail TCR Controller Trip TCR Emergency Switch Trip	TCR CONTROL FAILED/NORMAL	Trip
4	DISPLAY ON REACTOR CONTROL PANEL	Winding Temperature High – Stage-2	TCR #WDG.TEMP. HIGH ST 2 / NORMAL	Trip
5	DISPLAY ON REACTOR CONTROL PANEL	Winding Temperature High – Stage -1	TCR #WDG.TEMP. HIGH ST 1 / NORMAL	Non-Trip
6	DISPLAY ON REACTOR CONTROL PANEL	Buchholz Stage 1	TCR #BUCHHOLZ ST 1 ALARM / NORMAL	Non-Trip
7	DISPLAY ON REACTOR CONTROL PANEL	TCR Control Room Temperature High	TCR CONTROL ROOM TEMP HIGH/NORMAL	Non-Trip
8	DISPLAY ON REACTOR CONTROL PANEL	Air Condition Fail Air Condition Chiller Water Level Low	TCR AIR CONDITION FAULTY/NORMAL	Non-Trip

Table 27-18: Thyristor Controlled Reactor Alarms *(Continued)*

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
9	DISPLAY ON REACTOR CONTROL PANEL	Oil Temperature high Stage-1	TCR #OIL TEMP. HIGH ST. 1 / NORMAL	Non-Trip
10	DISPLAY ON REACTOR CONTROL PANEL	Oil Temperature high Stage-2	TCR #OIL.TEMP. HIGH ST. 2 / NORMAL	Non-Trip
11	DISPLAY ON REACTOR CONTROL PANEL	Oil level low	TCR #OIL.LEVEL LOW/NORMAL	Non-Trip
12	DISPLAY ON REACTOR CONTROL PANEL	Primary Protection Disturbed / Fail 1. Loss of DC 2. Relay Self Supervision 3. PT Failure 4. Protection Disabled	TCR #PROT FAILED/NORMAL	Non-Trip
13	DISPLAY ON REACTOR CONTROL PANEL	Backup Protection Disturbed / Fail 1. Loss of DC 2. Relay Self Supervision 3. PT Failure 4. Protection Disabled		
14	DISPLAY ON REACTOR CONTROL PANEL	Protection DC Supply Failure		
15	DISPLAY ON REACTOR CONTROL PANEL	Protection on TEST Position		
16	DISPLAY ON REACTOR CONTROL PANEL	Primary Protection Lock Out Operated	TCR #PROT LOCKOUT OPERATE / NORMAL	Non-Trip
17	DISPLAY ON REACTOR CONTROL PANEL	Secondary Protection Lock Out Operated		

= Reactor Number;
Please refer Table # 27-14 for alarms applicable to Delta / Star Reactors.

Table 27-19: Shunt Capacitor Bank Alarms

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	DISPLAY ON CAPACITOR CONTROL PANEL	Set -1 Protection Operated a. Phase Over - Current (O/C) b. Ground Time Over-Current (O/C) c. Residual over voltage (O/V) d. System Under voltage(U/V) e. Neutral Current Unbalance f. Voltage Differential – Stage 1 g. Voltage Differential – Stage 2	CAPACITOR BANK # PROTECTION SET 1 OPERATED/NORMAL	Trip
2	DISPLAY ON CAPACITOR CONTROL PANEL	Set -2 Protection Operated a. Phase Over - Current (O/C) b. Ground Time Over-Current (O/C) c. Residual over voltage (O/V) d. System Under voltage(U/V) e. Neutral Current Unbalance f. Voltage Differential – Stage 1 g. Voltage Differential – Stage 2	CAPACITOR BANK # PROTECTION SET 2 OPERATED/NORMAL	
3	DISPLAY ON CAPACITOR CONTROL PANEL	Set -1 Protection Lockout Relay Operated	CAPACITOR BANK # PROTECTION SET 1 LOCKOUT / NORMAL	
4	DISPLAY ON CAPACITOR CONTROL PANEL	Set -2 Protection Lockout Relay Operated	CAPACITOR BANK # PROTECTION SET 2 LOCKOUT / NORMAL	
5	DISPLAY ON CAPACITOR CONTROL PANEL	Protection Stage -1 Failure 1. Relay Defective 2. PT Supply Failure 3. DC Supply Failure	CAPACITOR BANK # PROT SET 1 FAILED / NORMAL	Non-Trip
6	DISPLAY ON CAPACITOR CONTROL PANEL	Protection Stage -2 Failure 1. Relay Defective 2. PT Supply Failure 3. DC Supply Failure	CAPACITOR BANK # PROT SET -2 FAILED / NORMAL	Non-Trip
# = CAPACITOR BANK NUMBER				

Table 27-20: Grounding Transformer Alarms

No.	Control Panel Annunciator	Initiator	Alarm Received At ECC	Status
1	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Phase Over (O/C) Protection Operated	GRND TRAFO # O/C OPERATED/NORMAL	Trip
2	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Neutral O/C Protection Operated		
3	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Buchholz Gas (1st Stage)	GRND TRAFO # BUCHHOLZ STG -1 OPERATED/NORMAL	Non-Trip
4	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Buchholz Gas (2nd Stage)	GRND TRAFO # BUCHHOLZ STG -2 OPERATED/NORMAL	Trip
5	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Pressure Relief Device Trip	GRND TRAFO # PRESS DEVICE OPERATED/NORMAL	Trip
6	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Oil level low	GRND TRAFO # OIL LEVEL LOW/NORMAL	Non-Trip
7	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Primary Protection Lockout Relay Operated	GRND TRAFO # PROT OPERATED/NORMAL	Trip
8	FLAG ON AUX.RELAY /DISPLAY ON NUMERICAL RELAY	Secondary Protection Lockout Relay Operated		
9	FLAG ON AUX.RELAY	Protection DC Supply Failure	GRND TRAFO # PRESS DEVICE OPERATED/NORMAL	Non-Trip
# = Grounding Transformer Number				

Table 27-21: Static Var Compensator (SVC)

NO	Control Panel Annunciator	Initiator	Alarm Received At ECC	TYPE
1	AUXILIARY SUPPLY FAIL.	Auxiliary Supply	SVC AUXILIARY SUPPLY FAILED/NORMAL	Non-Trip
2	COOLING SYSTEM FAULTY	Cooler Supply Supervision	SVC COOLING SYSTEM FAULTY/NORMAL	Non-Trip
3	REFERENCE VOLTAGE UP/DOWN	Regulation Control	SVC REF VOLTAGE UP/DOWN	Non-Trip
4	READY TO START	Ready To Start	RDY START	Non-Trip
5	URGENT ALARMS (GROUPED)	URGENT ALARMS (GROUPED)	URGENT ALARMS	Trip
6	NON-URGENT ALARM (GROUPED)	NON-URGENT ALARM (GROUPED)	NON-URGENT ALARMS	Non-Trip

Table 27-22: Circuit Switcher

NO	Control Panel Annunciator	Initiator	Alarm Received At ECC	TYPE
1	FLAG ON RELAYS	1. Low Gas Pressure 2. Trip Coil Failure 3. Trip Coil Control Supply Failure 4. Loss of AC / DC Supply	CIRCUIT SWITCHER FAULTY / NORMAL	Non-Trip

Table 27-23: Major SEC Customer's Station

NO	Control Panel Annunciator	Initiator	Alarm Received At ECC	TYPE
1	LOW POWER FACTOR	Low Power Factor (For each transmission line)	LOW PF	Non-Trip
2	MAX DEMAND EXCEEDED	Maximum Demand Exceeded	MAX DEM EXCD	
3	CUSTOMER TROUBLE	Customer Trouble Alarm (Combination of protection alarms on the customer side interface points)	CUST TRB	
4	GROUND SWITCH CLOSE	Ground Switch Status Indicator	GRD SW OPEN/CLOSE	
5	METER FAULT	Solid State Energy Meters, AC Power-Demand Meters and Power Factor Meters	Meter Fault	

Table 27-24: Command And Back Indications

NO	ALARM RECEIVED AT ECC	TEXT	TYPE
1	***BREAKER	ON/OFF	C-I
2	***ISOLATOR	ON/OFF	C-I
3	***E/SW	ON/OFF	I
4	*** 1ST PROTECTION	IN/OUT	C-I
5	*** 2ND PROTECTION	IN/OUT	C-I
6	*** AUTO REC	IN/OUT	C-I
7	*** AUTO REC RESET	RESET/NORMAL	C-I
8	*** REV.POWER	IN/OUT	C-I
9	***REMOTE OFF	IN/OUT	C-I
10	***V.SSYNCR	IN/OUT	C-I
11	***P.W.PROT	IN/OUT	C-I
12	***O/C PROT (B/S & B/C)	IN/OUT	C-I
13	***TOP	ON/OFF	C-I
14	***INTERLOCK	ON/OFF	I
15	***U/F PROT	IN/OUT	C-I
16	***OVER VOLT RELAY	IN/OUT	C-I
17	***ABTS	ON/OFF	C-I
18	###BB PROT	IN/OUT	C-I
19	###AUTO SW EQ	IN/OUT	C-I
20	BB CHK PROT	IN/OUT	C-I
21	===ACSE	IN/OUT	C-I
22	===TOP	IN/OUT	C-I
23	+++AVC	REMOTE/MANUAL	C-I
24	+++AVC	FOLLOWER / INDEPENDENT	C-I
25	+++AVC ECC AUTO	ON/OFF	C-I
26	+++AVC ECC MANUAL	ON/OFF	C-I

Table 27-24: Command And Back Indications *(Continued)*

NO	ALARM RECEIVED AT ECC	TEXT	TYPE
27	+++INDEPENDENT	ON/OFF	C-I
28	+++MASTER	ON/OFF	C-I
29	+++FOLLOWER	ON/OFF	C-I
30	+++AUTOMATIC	ON/OFF	C-I
31	+++MANUAL	ON/OFF	C-I
32	+++LOCAL	ON/OFF	C-I
33	+++TAP	RAISE/LOWER	C-I
34	+++LOAD SHEDDING 0%	ON/OFF	C-I
35	+++LOAD SHEDDING 1%	ON/OFF	C-I
36	+++LOAD SHEDDING 3%	ON/OFF	C-I
37	+++LOAD SHEDDING 5%	ON/OFF	C-I
38	+++LOAD SHEDDING 9%	ON/OFF	C-I
39	+++UNDER FREQ RLY	IN/OUT	C-I
40	+++UNDER FREQ RLY	IN/OUT	C-I
41	xxx SVC	START/STOP	C-I
42	xxx SVC REFERENCE VOLTAGE	RAISE/LOWER	C-I
43	xxx SVC EMERGENCY STOP	EMERGENCY STOP	C-I
44	xxx SVC COOLING	START/STOP	C-I
45	yyy TCR	START/STOP	C-I
46	yyy TCR REFERENCE VOLTAGE	RAISE/LOWER	C-I
47	yyy TCR MODE SELECT	MVAR/KV	C-I
48	www CAPACITOR BANK DISCONNECTOR SWITCH	OPEN/CLOSE	C-I
+++ = Transformer No. xxx = SVC No. yyy = TCR No. www = Capacitor Bank No.			

TABLE 27-25 Analog Measurand Value Acquisition at Local Control Panel and ECC

No	DATA SOURCE	MW	MVAR	MWH	MVARH	TAP	Volts	Amps
1	11 - 34.5 kV Feeder	-	-	-	-	-	√	√ (Each Phase)
2	69 - 380 kV Feeder	√	√	-	-	-	√	-
3	Power Transformer	√ (LV Side)	√ (LV Side)	-	-	√ (Position)	-	-
4	Static Var Compensator (SVC)	-	√	-	-	√ (Percent)	√ (& Ref Voltage)	-
5	Capacitor Bank	-	√	-	-	-	-	-
6	Thyristor Controlled Reactor (TCR)	-	√	-	-	√ (Percent)	√ (High Side)	√
7	Shunt Reactor	-	√	-	-	-	-	-
8	Bus Section	-	-	-	-	-	√ (Each Section)	-
9	Other Power Grids and SWCC Inter-Tie (Energy Metering)	-	-	√ 1.Gross (Export/Import) 2. Import 3. Export	√ 1. Gross (Export/Import) 2. Import 3. Export	-	-	-
10	Major SEC Customers (Energy Metering)	-	-	√ (Gross)	√ (Gross)	-	-	-

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